REVERSIBLE REACTIONS

Not all chemical reactions proceed to completion. In most reactions two or more substances react to form products which themselves react to give back the original substances. Thus A and B may react to form C and D which react together to reform A and B.

$$A + B \longrightarrow C + D \qquad \text{(Forward reaction)}$$

$$A + B \longleftarrow C + D \qquad \text{(Reverse reaction)}$$

A reaction which can go in the forward and backward direction simultaneously is called a Reversible reaction. Such a reaction is represented by writing a pair of arrows between the reactants and products.

$$A + B \rightleftharpoons C + D$$

The arrow pointing right indicates the forward reaction, while that pointing left shows the reverse reaction.

Some Examples of Reversible Reactions

A few common examples of reversible reactions are listed below :

$$2NO_2(g) \rightleftharpoons N_2O_4(g)$$

NATURE OF CHEMICAL EQUILIBRIUM : ITS DEFINITION

Let us consider the reaction

$$A + B \rightleftharpoons C + D$$

If we start with A and B in a closed vessel, the forward reaction proceeds to form C and D. The concentrations of A and B decrease and those of C and D increase continuously. As a result the rate of forward reaction also decreases and the rate of the reverse reaction increases. Eventually, the rate of the two opposing reactions equals and the system attains a *state of equilibrium*. Thus Chemical equilibrium may be defined as: **the state of a reversible reaction when the two opposing reactions occur at the same rate and the concentrations of reactants and products do not change with time.**

Furthermore, the true equilibrium of a reaction can be attained from both sides. Thus the equilibrium concentrations of the reactants and products are the same whether we start with A and B, or C and D.

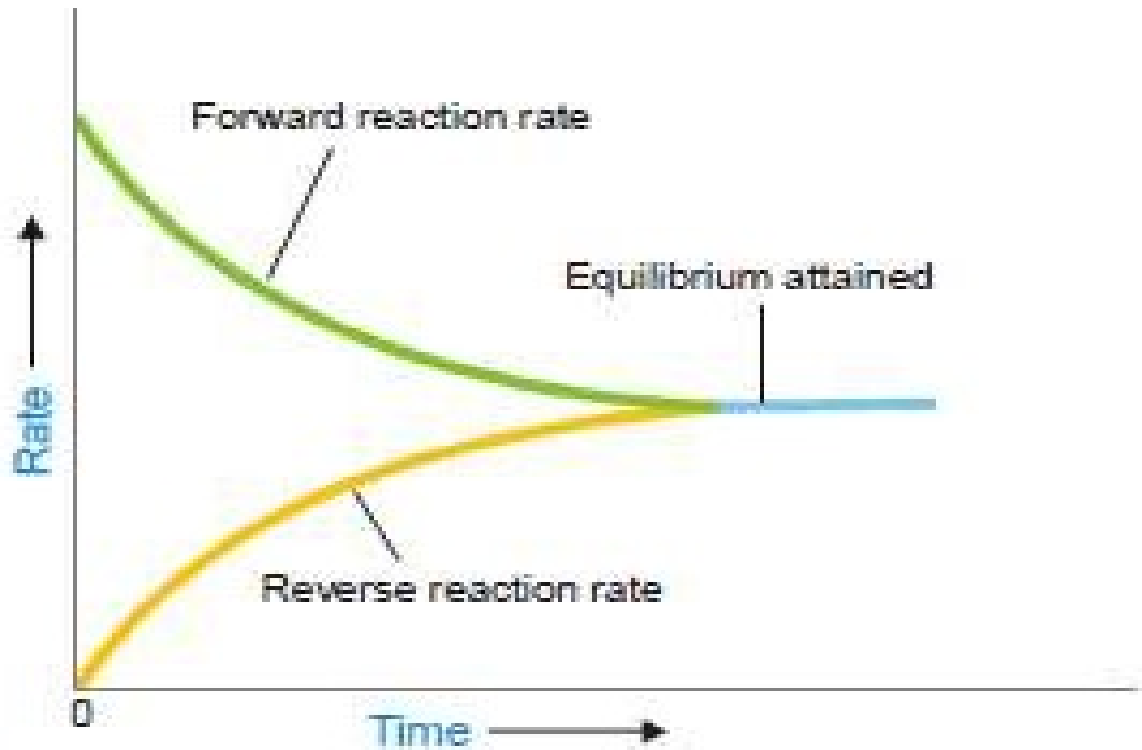


Figure 17.1
At equilibrium the forward reaction rate equals the reverse reaction rate.

Chemical Equilibrium is Dynamic Equilibrium

We have shown above that as the reaction $A + B \rightleftharpoons C + D$ attains equilibrium, the concentrations of A and B, as also of C and D remain constant with time. Apparently it appears that the equilibrium is dead. But it is not so. The equilibrium is dynamic. Actually, the forward and the reverse reactions are taking place at equilibrium but the concentrations remain unchanged.

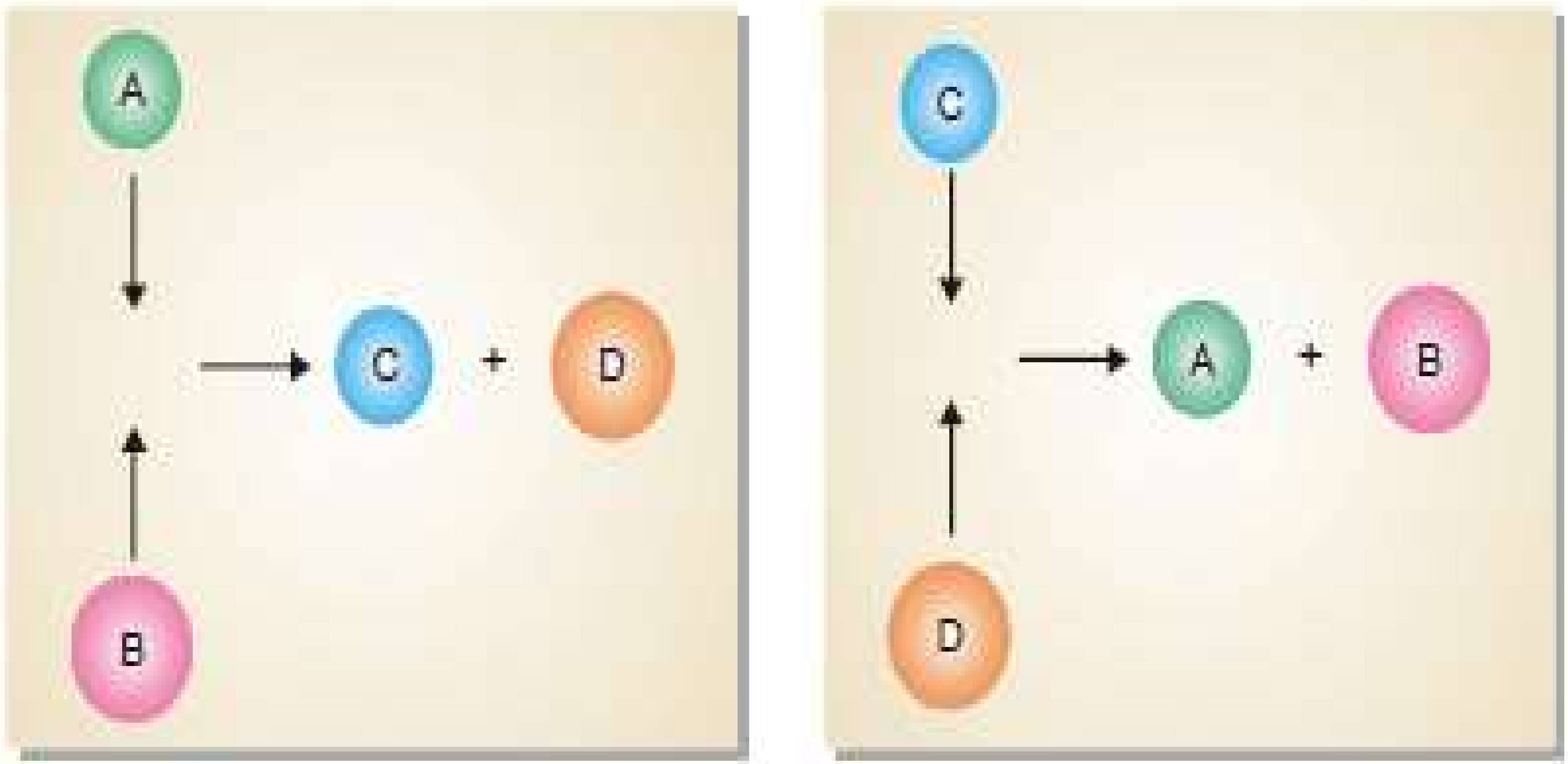


Figure 17.2
Molecules of A and B colliding to give C and D, and those of C and D colliding to give A and B.

The dynamic nature of chemical equilibrium can be easily understood on the basis of the kinetic molecular model. The molecules of A and B in the equilibrium mixture collide with each other to form C and D. Likewise C and D collide to give back A and B. The collisions of molecules in a closed system is a ceaseless phenomenon. Therefore collisions of A and B giving C and D (Forward reaction) and collisions of C and D giving back A and B (reverse reaction) continue to occur even at equilibrium, while concentrations remain unchanged.

LAW OF MASS ACTION

Two Norwegian chemists, Guldberg and Waage, studied experimentally a large number of equilibrium reactions. In 1864, they postulated a generalisation called the **Law of Mass action.** It states that : **the rate of a chemical reaction is proportional to the active masses of the reactants.**

By the term 'active mass' is meant the molar concentration *i.e.*, number of moles per litre. It is expressed by enclosing the formula of the substance in square brackets. Thus a gas mixture containing 1 g of hydrogen (H_2) and 127 g of iodine (I_2) per litre has the concentrations

$$[H_2] = 0.5 \text{ and } [I_2] = 0.5$$

EQUILIBRIUM CONSTANT : EQUILIBRIUM LAW

Let us consider a general reaction

$$A + B \rightleftharpoons C + D$$

and let [A], [B], [C] and [D] represent the molar concentrations of A, B, C and D at the equilibrium point. According to the Law of Mass action.

$$\text{Rate of forward reaction} \propto [A][B] = k_1 [A][B]$$

$$\text{Rate of reverse reaction} \propto [C][D] = k_2 [C][D]$$

where k_1 and k_2 are rate constants for the forward and reverse reactions.

At equilibrium, rate of forward reaction = rate of reverse reaction.

Therefore,

$$k_1[A][B] = k_2[C][D]$$

or $$\frac{k_1}{k_2} = \frac{[C][D]}{[A][B]} \qquad ...(1)$$

At any specific temperature k_1/k_2 is constant since both k_1 and k_2 are constants. The ratio k_1/k_2 is called Equilibrium constant and is represented by the symbol K_c, or simply k. The subscript 'c' indicates that the value is in terms of concentrations of reactants and products. The equation (1) may be written as

Equilibrium constant

$$k_c = \frac{[C][D]}{[A][B]}$$

Product concentrations

Reactant concentrations

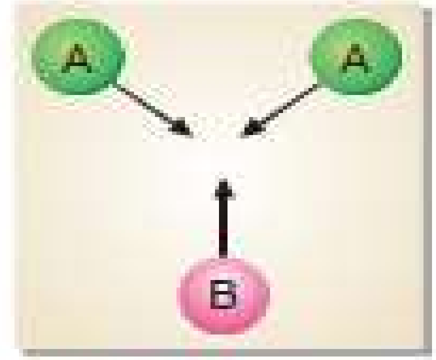


Figure 17.8

This equation is known as the **Equilibrium constant expression or Equilibrium law.**

Consider the reaction

Coefficient of A

$$2\,A \rightleftharpoons C + D$$

Here, the forward reaction is dependent on the collisions of each of two A molecules. Therefore, for writing the equilibrium expression, each molecule is regarded as a separate entity *i.e.*,

$$A + A \rightleftharpoons C + D$$

Then the equilibrium constant expression is

$$k_c = \frac{[C][D]}{[A][A]} = \frac{[C][D]}{[A]^2}$$

Power equal to coefficent of A

As a general rule, if there are two or more molecules of the same substance in the chemical equation, **its concentration is raised to the power equal to the numerical coefficient of the substance in the equation.**

Equilibrium Constant Expression for a Reaction in General Terms

The general reaction may be written as

$$aA + bB \rightleftharpoons cC + dD$$

where a, b, c and d are numerical quotients of the substance, A, B, C and D respectively. The equilibrium constant expression is

$$K_c = \frac{[C]^c\,[D]^d}{[A]^a\,[B]^b}$$

where K_c is the Equilibrium constant. The general definition of the equilibrium constant may thus be stated as : the product of the equilibrium concentrations of the products divided by the product of the equilibrium concentrations of the reactants, with each concentration term raised to a power equal to the coefficient of the substance in the balanced equation.

How to Write the Equilibrium Constant Expression?

(1) Write the balanced chemical equation for the equilibrium reaction. By convention, the substances on the left of the equation are called the **reactants** and those on the right, the **products.**

(2) Write the product of concentrations of the 'products' and raise the concentrations of each substance to the power of its numerical quotient in the balanced equation.

(3) Write the product of concentrations of 'reactants' and raise the concentration of each substance to the power of its numerical quotient in the balanced equation.

(4) Write the equilibrium expression by placing the **product concentrations in the numerator and reactant concentrations in the denominator.** That is,

$$K_c = \frac{\text{Product of concentrations of 'products' form Step (2)}}{\text{Product of concentrations of 'reactants' form Step (3)}}$$

SOLVED PROBLEM 1. Give the equilibrium constant expression for the reaction

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

SOLUTION

(1) The equation is already balanced. The numerical quotient of H_2 is 3 and of NH_3 is 2.

(2) The concentration of the 'product' NH_3 is $[NH_3]^2$

(3) The product of concnentrations of the reactants is $[N_2]\,[H_2]^3$

(4) Therefore, the equilibrium constant expression is

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

SOLVED PROBLEM 2. Write the equilibrium constant expression for the reaction

$$N_2O_5(g) \rightleftharpoons NO_2(g) + O_2\,(g)$$

SOLUTION

(1) The equation as written is not balanced. Balancing yields

$$2N_2O_5 \rightleftharpoons 4NO_2(g) + O_2$$

(2) The quotient of the product NO_2 is 4 and of the reactant N_2O_5 it is 2.

(3) The product of the concentrations of products is

$$[NO_2]^4\,[O_2]$$

(4) The concentration of the reactant is

$$[N_2O_5]^2$$

(5) The equilibrium constant expression can be written as

$$K_c = \frac{[NO_2]^4\,[O_2]}{[N_2O_5]^2}$$

UNITS OF EQUILIBRIUM CONSTANT

In the equilibrium expression for a particular reaction, the concentrations are given in units of moles/litre or mol/L, and the partial pressure are given in atmospheres (atm). The units of K_c and K_p depend on the specific reaction.

(1) When the total number of moles of reactants and products are equal

In the equilibrium expression of these reactions, the concentration or pressure terms in the numerator and denominator exactly cancel out. Thus K_c or K_p for such a reaction is without units.

Taking example of the reaction .

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

$$K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{(mol/L)^2}{(mol/L)(mol/L)} \quad \text{(No units)}$$

$$K_p = \frac{(P_{HI})^2}{(P_{H_2})(P_{I_2})} = \frac{(atm)^2}{(atm)(atm)} \quad \text{(No units)}$$

(2) When the total number of moles of the reactants and products are unequal

In such reactions K_c will have units $(\text{mol/litre})^n$ and K_p will have units $(\text{atm})^n$ where n is equal to the total number of moles of products minus the total number of moles of reactants. Thus for the reaction

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

$$K_c = \frac{[NO_2]^2}{[N_2O_4]} = \frac{(\text{mol/L})^2}{(\text{mol/L})} = \text{mol/L}$$

$$K_p = \frac{(p_{NO_2})^2}{(P_{N_2O_4})} = \frac{(\text{atm})^2}{(\text{atm})} = \text{atm}$$

Thus units for K_c are mol/L and for K_p units are atm. For the reaction

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

the units for K_c and K_p may be found as follows

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{(\text{mol/L})^2}{(\text{mol/L})(\text{mol/L})^3} = \text{mol}^{-2}\,\text{L}^2$$

$$K_p = \frac{(p_{NH_3})^2}{(p_{N_2})(p_{H_2})^3} = \frac{(\text{atm})^2}{(\text{atm})(\text{atm})^3} = \text{atm}^{-2}$$

Thus units of K_c are $\text{mol}^{-2}\,\text{L}^2$ and units of K_p are atm^{-2}.

It may be noted, however, that the units are often omitted when equilibrium constants are listed in tables.

HETEROGENEOUS EQUILIBRIA

So far we have discussed chemical equilibria in which all reactants and products are either gases or liquids. Such equilibria in which the reactants and products are not all in the same phase, are called **heterogeneous equilibria.** The decomposition of calcium carbonate upon heating to form calcium oxide and carbon dioxide is an example of heterogeneous equilibrium. If the reaction is carried in a closed vessel, the following equilibrium is established.

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$$

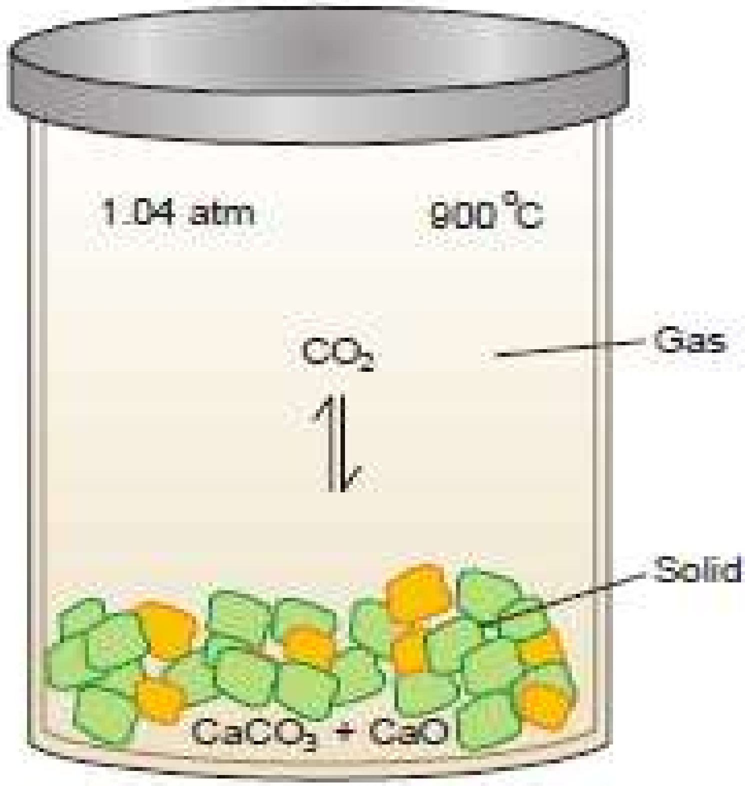


Figure 17.10
When solid $CaCO_3$ is heated in a closed vessel at 900 °C, equilibrium is established when the pressure of CO_2 reaches 1.04 atm.

What is Equilibrium?

This is not Equilibrium?

HOMOGENEOUS EQUILIBRIA

Chemical equilibrium in which all the reactants are in the same phase, are called homogeneous equilibria. For example in the synthesis of ammonia from nitrogen and hydrogen all reactant present in the same gaseous phase, so it is a homogeneous equilibria.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

The Relationship between K_c and K_p

- Consider the reaction: $2SO_{2(g)} + O_{2(g)} \rightleftarrows 2SO_{3(g)}$
- $K_c = \dfrac{[SO]^2}{[SO_2]^2[O_2]}$ and $K_p = \dfrac{(P_{SO3})^2}{(P_{SO2})^2(P_{O2})}$
- Assuming ideal behavior,
- where PV = nRT and P = (n/V)RT = [M]RT
- and $P_{SO3} = [SO_3]RT$; $P_{SO2} = [SO_2]RT$; $P_{O2} = [O_2]RT$

$$K_p = \frac{[SO_3]^2(RT)^2}{[SO_2]^2(RT)^2[O_2](RT)} = \frac{[SO_3]^2}{[SO_2]^2[O_2]}(RT)^{-1} = K_c(RT)^{-1}$$

Relationship between K_c and K_p

- For reaction: $PCl_{5(g)} \rightarrow PCl_{3(g)} + Cl_{2(g)}$;

$$K_p = \frac{(P_{PCl3})(P_{Cl2})}{(P_{PCl5})} = \frac{[PCl_3](RT) \text{ x } [Cl_2](RT)}{[PCl_5](RT)}$$

$$= \frac{[PCl_3][Cl_2]}{[PCl_5]}(RT)^1 = K_C(RT)^1$$

Relationship between K_c and K_p

- In general, for reactions involving gases such that,
  - $a\mathrm{A} + b\mathrm{B} \rightleftarrows c\mathrm{C} + d\mathrm{D}$

    where A, B, C, and D are all gases, and a, b, c, and d are their respective coefficients,
  - $K_p = K_c(\mathrm{RT})^{\Delta n}$

    and $\Delta n = (c + d) - (a + b)$

  (In heterogeneous systems, only the coefficients of the gaseous species are counted.)

Le Châtelier’s Principle

- The *Le Châtelier's principle* states that:

  when factors that influence an equilibrium are altered, the equilibrium will shift to a new position that tends to minimize those changes.

- Factors that influence equilibrium:

  Concentration, temperature, and partial pressure (for gaseous)

The Effect of Changes in Concentration

- Consider the reaction: $N_{2(g)} + 3H_{2(g)} \rightleftarrows 2\ NH_{3(g)}$;

$$K_c = \frac{[NH]^2}{[N_2][H_2]^3}$$

- If $[N_2]$ and/or $[H_2]$ is increased, $Q_c < K_c$
- ➔ a net forward reaction will occur to reach new equilibrium position.
- If $[NH_3]$ is increased, $Q_c > K_c$, and a net reverse reaction will occur to come to new equilibrium position.

LE CHATELIER'S PRINCIPLE

In 1884, the French Chemist Henry Le Chatelier proposed a general principle which applies to all systems in equilibrium. This important principle called the **Le Chatelier's principle** may be stated as : **when a stress is applied on a system in equilibrium, the system tends to adjust itself so as to reduce the stress.**

There are three ways in which the stress can be caused on a chemical equilibrium:

(1) Changing the concentration of a reactant or product.

(2) Changing the pressure (or volume) of the system.

(3) Changing the temperature.

Thus when applied to a chemical reaction in equilibrium, Le Chatelier's principle can be stated as: **if a change in concentration, pressure or temperature is caused to a chemical reaction in equilibrium, the equilibrium will shift to the right or the left so as to minimise the change.**

Now we proceed to illustrate the above statement by taking examples of each type of stress.

EFFECT OF A CHANGE IN CONCENTRATION

We can restate Le Chatelier's principle for the special case of concentration changes: **when concentration of any of the reactants or products is changed, the equilibrium shifts in a direction so as to reduce the change in concentration that was made.**

Addition of an inert gas

Let us add an *inert gas* to an equilibrium mixture while the volume of the reaction vessel remains the same. The addition of the inert gas increases the total pressure but the partial pressures of the reactants and products are not changed. Thus **the addition of an inert gas has no effect on the position of the equilibrium.**

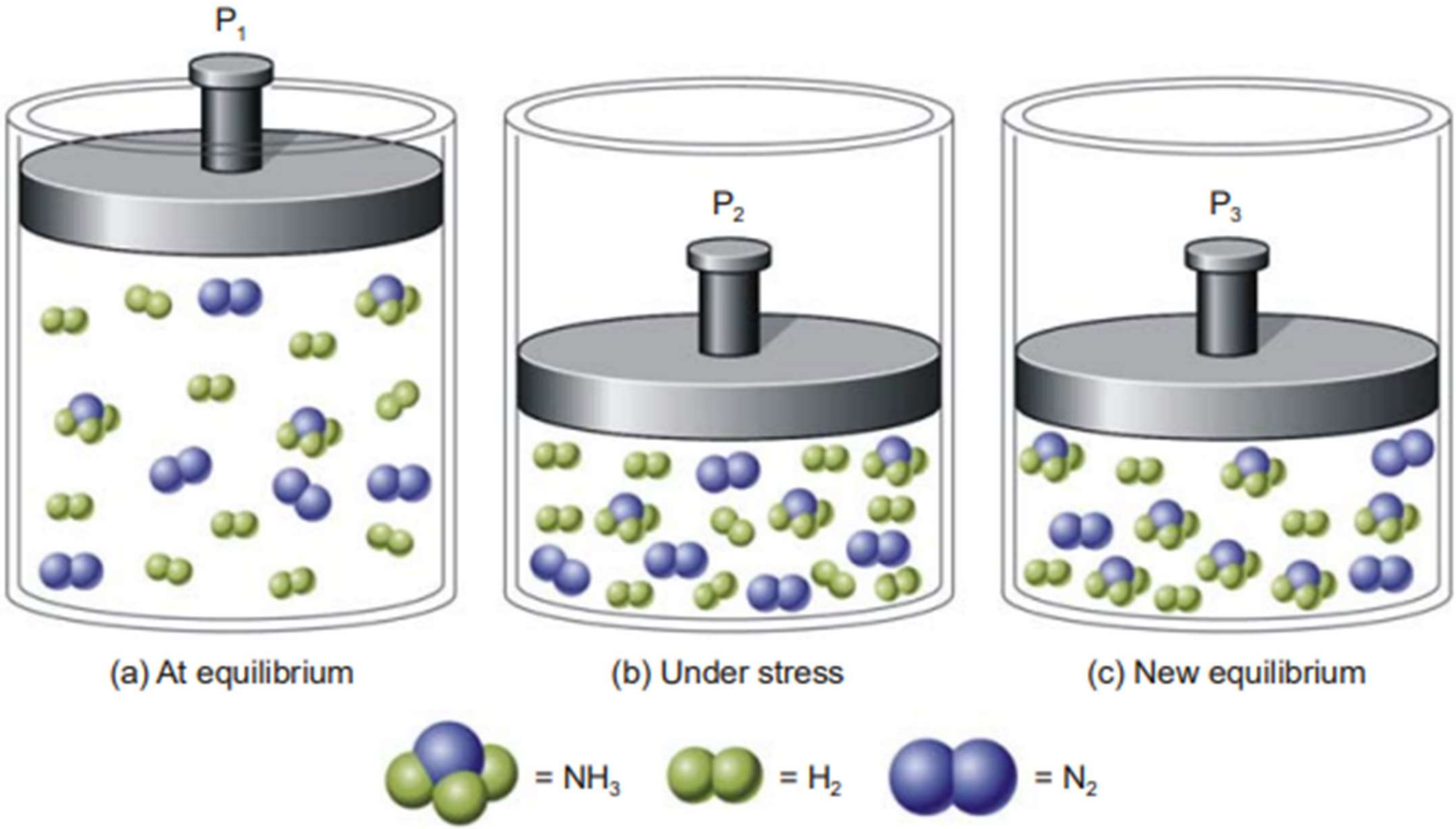


Figure 17.11

Illustratation of Le Chatelier's principle. (*a*) System at equilibrium with $10H_2$, $5N_2$, and $3NH_3$, for a total of 18 molecules. (*b*) The same molecules are forced into a smaller volume, creating a stress on the system. (*c*) Six H_2 and $2N_2$ have been converted to $4NH_3$. A new equilibrium has been established with $4H_2$, $3N_2$, and $7NH_3$, a total of 14 molecules. The stress is partially relieved by the reduction in the total number of molecules.

Effect of change of concentration on Ammonia Synthesis reaction

Let us illustrate the effect of change of concentration on a system at equilibrium by taking example of the ammonia synthesis reaction:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

When N_2 (or H_2) is added to the equilibrium already in existence (equilibrium I), the equilibrium will shift to the right so as to reduce the concentration of N_2 (Le Chatelier's principle). The concentration of NH_3 at the equilibrium II is more than at equilibrium I. The results in a particular case after the addition of one mole/litre are given below.

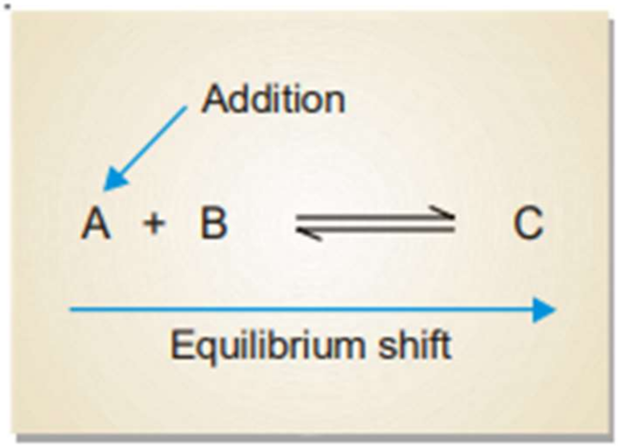
Equilibrium I		Equilibrium II
$[N_2] = 0.399$ M		$N_2 = 1.348$ M
$[H_2] = 1.197$ M	$\xrightarrow[\text{of } N_2 \text{ added}]{\text{1 mole/L}}$	$[H_2] = 1.044$ M
$[NH_3] = 0.202$ M		$[NH_3] = 0.304$ M

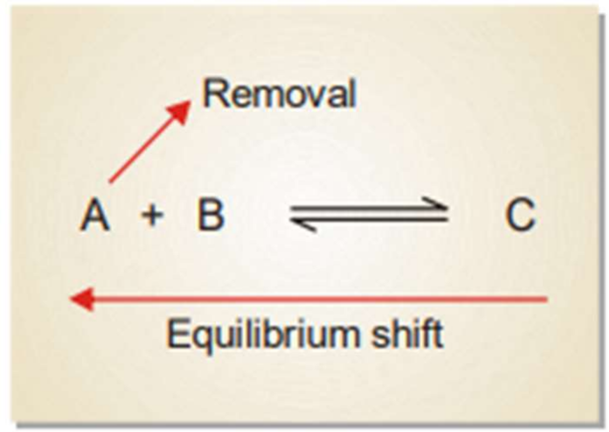
Obviously, the addition of N_2 (a reactant) increases the concentration of NH_3, while the concentration of H_2 decreases. Thus to have a better yield of NH_3, one of the reactants should be added in excess.

A change in the concentration of a reactant or product can be effected by the addition or removal of that species. Let us consider a general reaction

$$A + B \rightleftharpoons C$$

When a reactant, say, A is added at equilibrium, its concentration is increased. The forward reaction alone occurs momentarily. According to Le Chatelier's principle, a new equilibrium will be established so as to reduce the concentration of A. Thus the addition of A causes the equilibrium to shift to right. This increases the concentration (yield) of the product C.





Following the same line of argument, a decrease in the concentration of A by its removal from the equilibrium mixture, will be undone by shift to the equilibrium position to the left. This reduces the concentration (yield) of the product C.

EFFECT OF A CHANGE IN PRESSURE

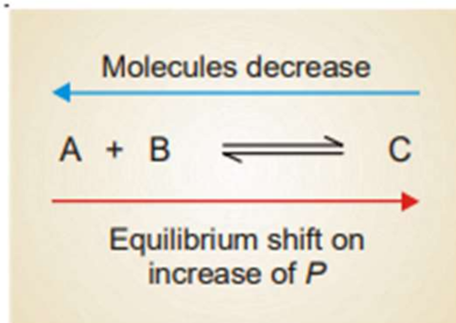
To predict the effect of a change of pressure, Le Chatelier's principle may be stated as : **when pressure is increased on a gaseous equilibrium reaction, the equilibrium will shift in a direction which tends to decrease the pressure.**

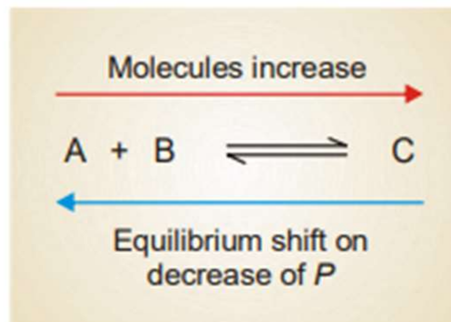
The pressure of a gaseous reaction at equilibrium is determined by the total number of molecules it contains. If the forward reaction proceeds by the reduction of molecules, it will be accompanied by a decrease of pressure of the system and *vice versa*.

Let us consider a reaction,

$$A + B \rightleftharpoons C$$

The combination of A and B produces a decrease of number of molecules while the decomposition of C into A and B results in the increase of molecules. Therefore, by the increase of pressure on the equilibrium it will shift to right and give more C. A decrease in pressure will cause the opposite effect. The equilibrium will shift to the left when C will decompose to form more of A and B.





The reactions in which the number of product molecules is equal to the number of reactant molecules,

e.g., $$H_2(g) + I_2(g) \rightleftharpoons 2HI$$

are unaffected by pressure changes. In such a case the system is unable to undo the increase or decrease of pressure.

In light of the above discussion, we can state a general rule to predict the effect of pressure changes on chemical equilibria.

The increase of pressure on a chemical equilibrium shifts it in that direction in which the number of molecules decreases and vice-versa. This rule is illustrated by the examples listed in Table 17.1.

EFFECT OF CHANGE OF TEMPERATURE

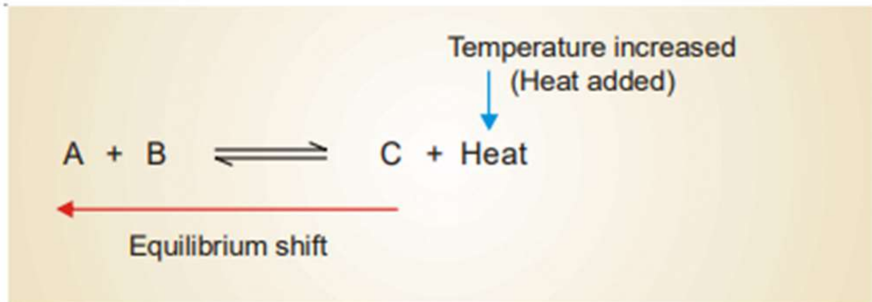
Chemical reactions consist of two opposing reactions. If the forward reaction proceeds by the evolution of heat (exothermic), the reverse reaction occurs by the absorption of heat (endothermic). Both these reactions take place at the same time and equilibrium exists between the two. If temperature of a reaction is raised, heat is added to the system. The equilibrium shifts in a direction in which heat is absorbed in an attempt to lower the temperature. Thus the effect of temperature on an equilibrium reaction can be easily predicted by the following version of the Le Chatelier's principle.

When temperature of a reaction is increased, the equilibrium shifts in a direction in which heat is absorbed.

Let us consider an exothermic reaction

$$A + B \rightleftharpoons C + \text{heat}$$

When the temperature of the system is increased, heat is supplied to it from outside. According to Le Chatelier's principle, the equilibrium will shift to the left which involves the absorption of heat. This would result in the increase of the concentration of the reactants A and B.

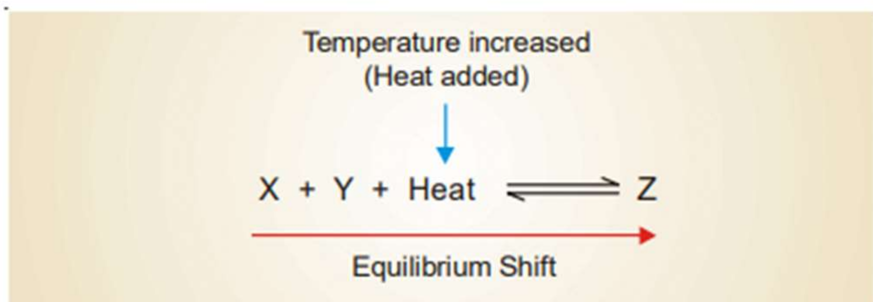


In an endothermic reaction

$$X + Y + \text{heat} \rightleftharpoons Z$$

the increase of temperature will shift the equilibrium to the right as it involves the absorption of heat. This increases the concentration of the product Z.

In general, we can say that **the increase of temperature favours the reverse change in an exothermic reaction and the forward change in an endothermic reaction.**



Formation of Ammonia from N_2 and H_2

The synthesis of ammonia from nitrogen and hydrogen is an exothermic reaction.

$$N_2 + 3H_2 \rightleftharpoons 2NH_3 + 22.2\,\text{kcal}$$

When the temperature of the system is raised, the equilibrium will shift from right-to-left which absorbs heat (*Le Chatelier's principle)* This results in the lower yield of ammonia. On the other hand, by lowering the temperature of the system, the equilibrium will shift to the right which evolves heat in an attempt to raise the temperature. This would increase the yield of ammonia. But with decreasing temperature, the rate of reaction is slowed down considerably and the equilibrium is reached slowly. Thus in the commercial production of ammonia, it is not feasible to use temperature much lower than 500°C. At lower temperature, even in the presence of a catalyst, the reaction proceeds too slowly to be practical.

Effects of Pressure Change on Equilibrium

- If the volume of a gas mixture is compressed, the overall gas pressure will increase. In which direction the equilibrium will shift in either direction depends on the reaction stoichiometry.

- However, there will be no effect to equilibrium if the total gas pressure is increased by adding an inert gas that is not part of the equilibrium system.

The Effect Temperature on Equilibrium

- Consider the following exothermic reaction:

$$N_{2(g)} + 3H_{2(g)} \rightleftarrows 2NH_{3(g)}; \qquad \Delta H^o = -92 \text{ kJ},$$

- The forward reaction produces heat => heat is a product.
- When heat is added to increase temperature, reverse reaction will take place to absorb the heat;
- If heat is removed to reduce temperature, a net forward reaction will occur to produce heat.
- Exothermic reactions favor low temperature conditions.

Chemical Equilibrium

- What is equilibrium?
- Relationship between K_c and K_p;
- Factors that affect equilibrium;
- Le Chatelier’s Principle